

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO2: Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)

Assessment Tool Weightage: 40%

QUESTIONS: 3

Duration: 1 Hour
Total Marks: 30

Annexure A

Dry Run Evaluation

Code of Conduct:

I M. PRASANNA VENKATESH (190573016) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



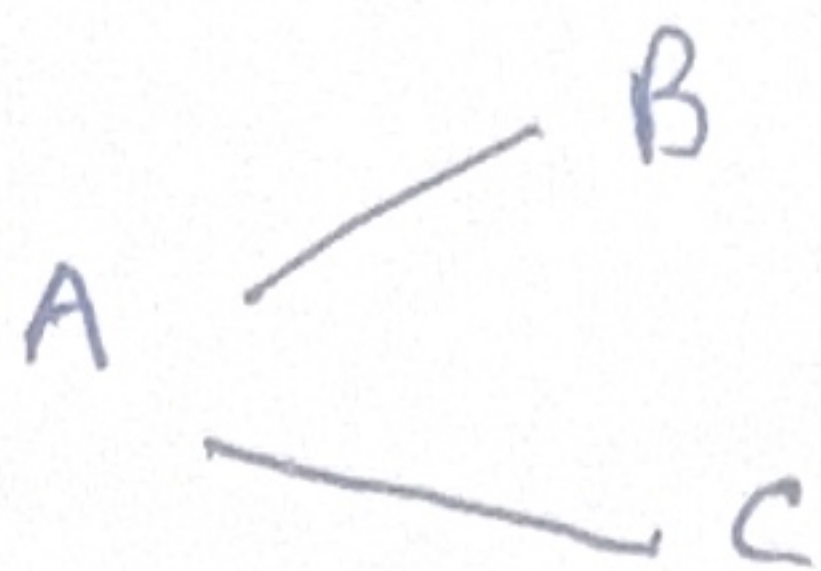
SEARCH WITH PARTIAL INFORMATION:

A robot must travel A to G. Initially, it knows only:

$$A \rightarrow B$$

$$A \rightarrow C$$

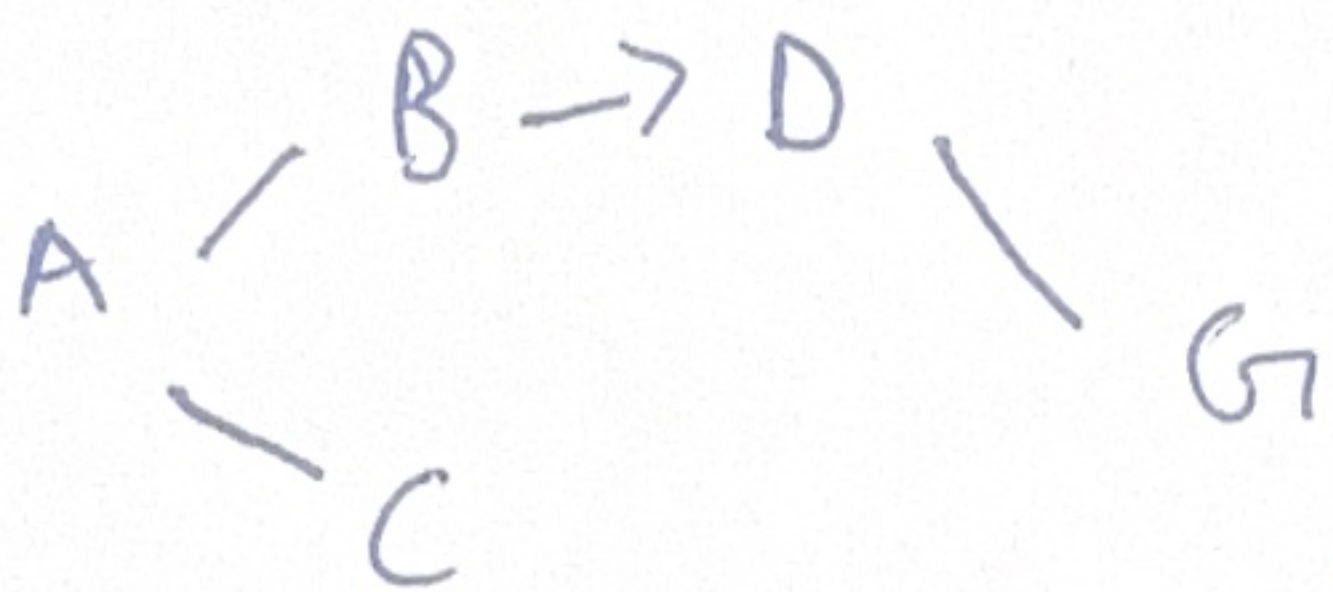
So the initial search space is



when the robot reaches B, it discovers:

$$B \rightarrow D$$

The search space becomes:



later, when the robot reaches D, it discovers:

$$D \rightarrow G$$

Now the complete path is

$$A \rightarrow B \rightarrow D \rightarrow G$$

CONCLUSION:

The search space grows as the robot discovers new information. This is example of search with partial information, where the robot plans while exploring.

22) BIDIRECTIONAL SEARCH

Given:

$A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow F$

Start = A

Goal = F

Bidirectional Search searches from both ends at the same time.

Forward Search.

$A \rightarrow B \rightarrow C$

Backward Search

$F \rightarrow E \rightarrow D$

The two searches meet at C/D depending on the expansion step. Using equal expansion levels.

Forward : $A \rightarrow B \rightarrow C$

Backward : $F \rightarrow E \rightarrow D$

The connecting path is:

$A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow F$

Advantage:

Instead of searching the entire path from one side, it searches from both directions, reducing the search effort.

Conclusion:

The searches meet around the middle of the graph, making Bidirectional Search faster than a one-direction search in suitable problems.

AGENT ARCHITECTURE - SELF DRIVING CAR:

SENSORS:

The sensors collect information from the environment:

- * Camera - Detects roads, vehicles and signs.
- * GPS - gives location.
- * LIDAR - detects objects and distances.
- * Speed sensor - measures vehicle speed.

Actuators:

The actuators perform actions:

- * Steering - Controls direction.
- * Brake - Slows / stops the car.
- * Accelerator - Controls speed.

Environment:

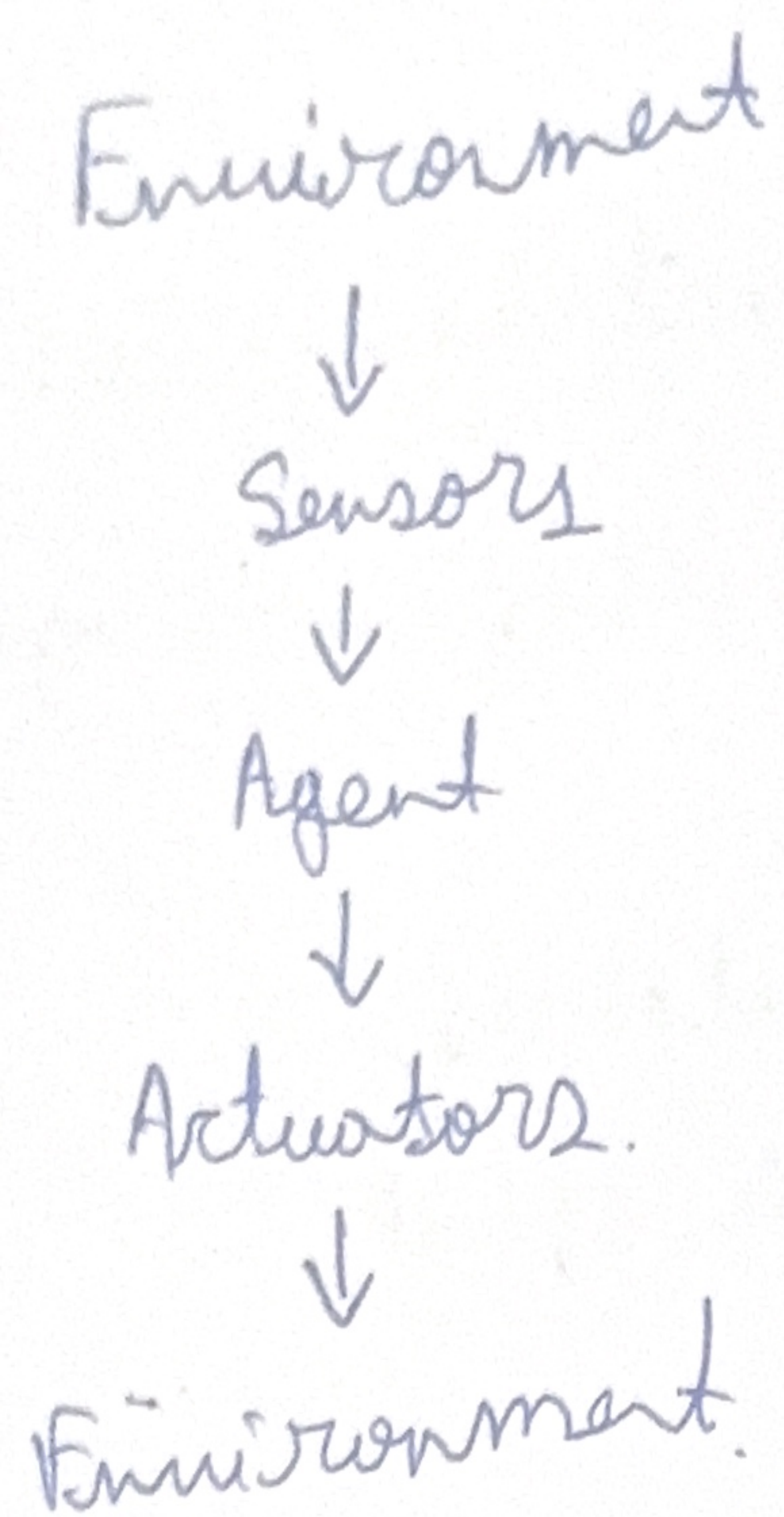
Roads, vehicles, pedestrians, traffic signals, buildings and weather.

Percepts:

Percepts are information received by the car, such as:

- * Vehicle detected.
- * Current location.
- * Distance to obstacle.
- * Current Speed.
- * Traffic Signal Status.

Basic Agent Structure:



SUITABLE AGENT TYPE:

A Model-Based Utility Agent is suitable because the car must:

- * Maintain information about its surroundings.
- * Make decisions using sensor data.
- * Consider safety and efficiency.
- * Choose the best action.

CONCLUSION:

A model-based utility agent is suitable for safe and intelligent self-driving.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

$$19 + 19 + 23 + 18 + 12$$

91